🔑 **Answer Key — Teacher Copy · Lesson 2-3**

MSTAR PRACTICE WORKSHEET

Lesson 2-3: Describe the Data Using the Median

Grade 6 Mathematics · Standard 6.DS.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

Find the mean of: 10, 14, 8, 12, 16

A. 10

B. 12

C. 14

D. 60

Sum = 10 + 14 + 8 + 12 + 16 = 60. Mean = 60 ÷ 5 = 12.

Question 2 SELECTED RESPONSE — CORRECT: B

Find the median of: 3, 7, 2, 9, 5

A. 3

B. 5

C. 5.2

D. 7

Ordered: 2, 3, 5, 7, 9. The middle value is 5.

Question 3 SELECTED RESPONSE — CORRECT: B

Find the mode of: 4, 7, 4, 9, 2, 4, 8

A 2

B 4

C 7

D 9

4 appears three times — more than any other value. Mode = 4.

Question 4 SELECTED RESPONSE — CORRECT: B

Find the mean of: 6, 8, 10, 12, 14

A 8

B 10

C 12

D 50

Sum = 6 + 8 + 10 + 12 + 14 = 50. Mean = 50 ÷ 5 = 10.

Question 5 SELECTED RESPONSE — CORRECT: B

Data set: 4, 6, 6, 8, 10, 12. What is the median?

A 6

B 7

C 7.67

D 8

With 6 values, the median is the average of the 3rd and 4th values: (6 + 8) ÷ 2 = 7.

Question 6 SELECTED RESPONSE — CORRECT: A

A player scored 4, 8, 6, 10, and 2 points across five quarters. What is the MEAN number of points?

- (A) 6
- (B) 8
- (C) 10
- (D) 30

Add the points: 4 + 8 + 6 + 10 + 2 = 30. Divide by 5 quarters: 30 ÷ 5 = 6.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

Find the median of this data set: 12, 7, 15, 9, 20, 11, 8.

- (A) 9
- (B) 11
- (C) 12
- (D) 15

Ordered: 7, 8, 9, 11, 12, 15, 20. There are 7 values, so the median is the 4th one: 11.

- **A:** That is the middle of the list as written. Order the values first: 7, 8, 9, 11, 12, 15, 20.
- **C:** That is the first value listed, not the middle one. Sort the seven values and take the 4th.
- **D:** That is the 4th value before sorting. The median is the 4th value AFTER sorting.

Part B — correct answer: C

Why must the data be put in order before finding the median?

- (A) Ordering makes the numbers easier to add.
- (B) The median is always the largest value.
- (C) The median is the middle value by position in the ordered list, not by where it was written.
- (D) Ordering changes the median to a whole number.

The median is defined by position in an ordered list. Taking the middle of an unordered list picks an arbitrary value with no meaning.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: The Unordered Middle

> Kai found the median of 14, 22, 9, 30, 18. He wrote: 'The middle number written down is 9, so the median is 9.'

Explain Kai's misconception and give the correct median.

Model answer: Ordered: 9, 14, 18, 22, 30. With 5 values the median is the 3rd, which is 18. Kai took the middle of the list as written; the median is the middle of the list in order.

2 points	Student orders the set as 9, 14, 18, 22, 30, identifies 18 as the 3rd of 5 values, and explains that the median is the middle of the ORDERED list.
1 point	Student gives 18 without explaining why ordering is required.
0 points	Student agrees with Kai or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A basketball player scored 15, 18, 16, 42, and 14 points in 5 games. Find the mean and median. Which better represents the player's typical game? Explain how the outlier affects each measure.

Model answer: The five scores add to 105, so the mean is 105 ÷ 5 = 21, while the ordered list 14, 15, 16, 18, 42 gives a median of 16. The 42-point game is an outlier that drags the mean up above four of the five games, but the median only cares about position, so 16 better represents a typical game.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.